



Fig. 10: CapSense image

voltage output of the accelerometer is routed to an ADC in the PSoC. This digital value is now a measure of acceleration detected by the accelerometer. If acceleration exceeds the maximum value, it signals the start of a new wave event.

Firmware delays are for a fixed period of time to start the first character in the proper position of the wave. PSoCRocks string is converted into a rasterised display data array before the start of the wave. Each column of the rasterised data array is sequentially output after a fixed delay.

Delay periods may be adjusted in the firmware. For a particular character in the wave, a look-up table determines which LEDs to turn on. When the board is waved back and forth rapidly, due to persistence of vision, the user-defined image appears to float in the air. Multiple strings may be configured with each string being displayed for an adjustable number of wave cycles.

PSoCRocks application displays a rasterised image of the user-defined message ("PSoCRocks!") using persistence of vision. Waving the circuit board back and forth quickly in the plane of the board generates the message.

Bubble levels application using PSoC board. A bubble level displays the degree of horizontal deviation using a bubble of air in a glass via level. In firmware project, a level is emulated by the accelerometer on the board. In bubble-level detection, LEDs display the tilt of the board along its length (corresponding to x axis of the accelerometer).

The accelerometer is always

under the influence of gravity. When the board is placed flat on the ground, static acceleration due to gravity along x axis of the accelerometer is zero. In this position, two LEDs in the centre of the LED display are turned on. Because the board is tilted in one direction, force of gravity along that axis increases. This increases static acceleration detected by the accelerometer.

The x axis voltage output of the accelerometer is fed into the ADC in the PSoC. The ADC converts this voltage into digital count, which is the measure of static acceleration detected by the accelerometer. PSoC determines the tilt of the board and lights up LEDs indicating the tilt. If the board is further tilted in one direction, the LED moves accordingly to the extreme of the LED display.

CapSense inputs using PSoC board. Capacitive sensing determines the presence of a conductive element such as a finger on a capacitive sensor incorporated on the PCB. The kit consists of CapSense sensors in the form of a slider. Size of the sensors and their positions are designed such that when a finger is placed on any part of the slider, at least three sensors are activated (it detects the presence of the finger by a change in its capacitance value).

CapSense component provides APIs, which report the centroid (relative position) of the finger on the slider based on active sensors. The firmware then lights up the LED corresponding to this centroid position.

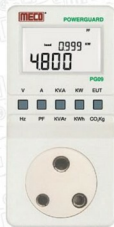
Conclusion

A PSoC is a combination of an MCU and an ASIC. It provides an ease-of-use environment in electronic toy applications. Using a PSoC in these applications helps reduce the product cost (by reducing bills of material cost and design cycle time) and project cost (with PSoC Creator and PSoC Designer implementation). **EFY**

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